

## Reservoir fluid identification based on multi-head attention with UMAP

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### ABSTRACT

Gas logging curves obtained during drilling, along with conventional well logs, typically serve as foundational data for identifying reservoir fluids. Traditional methods for identifying layers containing oil or gas often suffer from limitations such as inadequate feature extraction, underutilization of differences in data points, and limited applicability. These limitations lead to reduced accuracy in predicting reservoir fluid segmentation. To address the challenge of spatiotemporal feature extraction in logging curves, this paper presents a fluid identification method based on deep learning networks and multivariate sequence classification. The method takes into account both the sequential variations in geological strata and the relationships among different curves for fluid prediction. Our proposed method utilizes a Transformer multi-head attention mechanism and follows a series of steps. First, an improved Symbolic Aggregate approXimation (SAX) is employed to assess the response characteristics of the logging curves to the target fluid and identify sensitive curves based on information similarity. Subsequently, the semi-supervised learning method UMAP is applied to identify layers with high interpretability. Hierarchical training is then employed to mitigate the impact of noise and high-dimensional nonlinearity on identification results. Finally, the processed vector curves are input into the attention module layer, where features of each layer are extracted in an encoding-decoding manner. This method has been tested and validated using gas logging data from 12 wells in a gas field offshore China. The overall prediction accuracy can exceed 94%, and the prediction accuracy for a single gas layer is over 88%. When compared to other machine learning approaches, such as K-means, decision tree, and XGBoost, this method exhibits superior identification accuracy and generalization ability, especially for unbalanced samples.

### 1. Introduction

In contemporary natural gas exploration and development, various methods such as seismic reflection wave analysis, chemical sampling data analysis, and logging curve identification are used to ascertain the distribution and content of natural gas in subterranean reservoirs. These methods also help interpret the properties of unknown fluids within target reservoirs.

Logging interpretation methods primarily fall into two categories. The first category involves mathematical statistics, wherein a template for fluid identification and lithology delineation is established through the construction of a crossplot diagram. This method also involves creating empirical formulas using core analysis data. The second category includes volumetric modeling and theoretical formulas, such as

porosity and saturation formulas. These methods enable the quantitative evaluation of reservoirs through impact equations and optimization methods (Zhang et al., 2020; Hou et al., 2022; He et al., 2023). However, challenges arise due to the limitations in on-site data collection for gas testing. Typically, only data for a limited number of oil, gas, and water layers are obtained from drilling samples, leading to an unbalanced distribution of categories. Consequently, the ability to accurately predict fluid types in regions with sparsely labeled data remains a persistent challenge in fluid identification tasks.

Conventional research methods often rely on a limited set of logging curves and empirical equations to interpret oil and gas layers. These methods derive various discriminant parameters and determine reservoir properties within specific numerical ranges. Examples include those based on indirect parameter optimization methods, such as the cross-

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plotting method (Xu et al., 2012; Zhang et al., 2012), the stratigraphic correlation index method (Ge et al., 2022), and methods utilizing parameters like lithology, seismic mechanics data, or carbon chemistry content (Xu et al., 2017). However, these approaches fail to fully leverage the diverse attributes of numerous logging curve features and often suffer from low regional applicability in real-world scenarios. For unconventional oil and gas reservoirs, such as shale oil and gas reservoirs, due to the complex composition of lithologic minerals, diverse occurrence patterns of oil and gas, strong heterogeneity, poor physical properties, and other characteristics, the established cross-sectional map or simplified volume model are not suitable (Soltanmohammadi et al., 2024). Additionally, the relationship between well data and actual core experimental data is often non-linear, and the established linear empirical formula has poor accuracy and generalization ability.

In the era of digital logging, machine learning methods offer a more automated approach for reservoirs with complex fluid distributions. Current fluid identification methods generally focus on either directly predicting the target reservoir's fluid category or indirectly assessing fluid properties through intermediate parameters like porosity and saturation. For example, by combining the Long Short-Term Memory (LSTM) network with the attention mechanism and inputting spatial coordinates alongside sedimentary landform information, Zhang et al. (2024) were able to predict the well log porosity curve at a designated location within the study area. Logging curves with depth as a sequence contain many variable dimensions, but the information that truly distinguishes the classification results usually only exists in a few curves, making it difficult to extract information directly from a large number of curves using traditional neural networks. Machine learning methods with strong non-linear fitting capabilities have rapidly advanced, especially with the rise of deep learning, achieving significant progress in fields such as image recognition, speech recognition, and natural language processing. For instance, in intelligent interpretation of geophysical well logging, the integration of rock property data and corresponding logging data using the K-means unsupervised clustering algorithm has successfully characterized reservoir sequences (Iraji et al., 2023).

Zhu et al. (2023) proposed that through K-means clustering and Adaboost methods, the K-means method could correspond to the hydrate enrichment degree, while the random forest method could establish an effective reservoir classification model. Additionally, techniques like Extreme Gradient Boosting (XGBoost) have been applied to identify fluid properties from logging data in low-permeability sandstone reservoirs, among other applications (Lou et al., 2024). In the face of small samples, transfer learning-based approaches transfer pre-trained models to fluid identification, such as reservoir identification methods combining convolutional mask attention networks and value-aware meta-transfer learning strategies (Chen et al., 2023).

To address these limitations, this study employs a time series SAX method (Malinowski et al., 2013). This method is combined with curve fluctuation information to construct logging curve importance evaluation indicators. The approach takes into account both the curve's response to each target layer and the distance similarity between curves. A new curve sensitivity analysis method is constructed using this approach, followed by a multi-head attention classifier with the UMAP (Uniform Manifold Approximation and Projection) algorithm. This combination aims to optimize the loss function for fluid identification.

In logging interpretation, various methods exist for screening sensitive curves. These include feature dimension reduction techniques (Jilinski, 2015; Wilkie et al., 2019), graphical segmentation methods (Yin et al., 2023), correlation coefficient-based approaches (Zhou et al., 2022), methods based on computed response-derived parameters (Ma et al., 2023), and machine learning techniques like XGBoost importance analysis (Yan, 2022). While these methods can capture the relationship or spatial distance between curves, they often fall short in handling nonlinear relationships or extracting trend information.

The relationship between curves in well logging models is intricate.

Even after identifying sensitive curves, machine learning models are essential for training and predicting data to meet practical requirements. Existing fluid classifiers often draw inspiration from image or natural language processing fields. Examples include density clustering-based nearest neighbor methods (Zhao et al., 2015), multiple decision tree-based interpretation methods (He et al., 2022), integrated multi-classifiers (Bai et al., 2021), convolutional neural networks (Imamverdiyev and Sukhostat, 2019), bidirectional long short-term memory network-based identification methods (Zhou et al., 2021), and reservoir classification model mixed with multiple sub-algorithms (Liu et al., 2023). However, these methods primarily focus on point-to-point input-output relationships and often employ simple oversampling techniques for handling unbalanced samples. This results in a lack of coherence between local relationships in reservoirs and curves, making the samples susceptible to disturbances and reducing effectiveness.

To enhance the accuracy of fluid identification and its applicability, this paper introduces a fluid identification algorithm that combines UMAP dimensionality reduction clustering with a Transformer multi-head attention mechanism encoder (Vaswani et al., 2023). Transformers offer interpretability through their self-attention mechanisms, allowing users to understand which parts of the input sequence contribute most to the model's predictions (Zhang et al., 2024). Operating in a semi-supervised fashion, the method extracts layers with high interpretability. It utilizes position encoding and an attention model to scrutinize curve information, thereby optimizing classification performance for unbalanced samples. As a result, this approach provides distinct advantages over existing fluid identification methods. Overall, this paper aims to propose a method to characterize the reservoir fluid types more accurately.

The proposed method introduces several innovative aspects, including: 1) Curve sensitivity detection based on dimensionality reduction and word frequency: The method constructs indicators to evaluate the importance of logging curves and measures the similarity between curves. 2) A novel approach to stratigraphic division: Our method utilizes UMAP for dimensionality reduction, demonstrating its effectiveness in capturing intricate relationships within high-dimensional data. 3) Integration of Attention Mechanism: We have incorporated an attention mechanism into the model to focus on key features and patterns in the data. Through sequence input, we can not only fully utilize the response value of the logging curve corresponding to depth, but also consider the trend and correlation of the logging curve with depth. This approach helps avoid the impact of reservoir longitudinal heterogeneity to the greatest extent.

This paper is organized as follows: Firstly, the geological setting and the data are introduced. Next, the workflow of the proposed methodology is provided. Following that, technical details of two main procedures: SAX of logging curves and the Transformer Classification Method Integrated with UMAP are described in detail. Finally, the method is validated through experiments.

## 2. The geological setting and dataset

### 2.1. Source of gas logging data

Associated gas is a byproduct generated during crude oil formation. The quantity of associated gas increases with lighter oil quality and higher maturity levels in the reservoir. This gas exists in two states: free and dissolved. When the drill bit penetrates the oil and gas reservoir, the reservoir fluid is carried upward by the drilling fluid. A subsequent drop in pressure and temperature triggers the expansion and precipitation of the natural gas. The degasser then isolates this natural gas from the drilling fluid. Its composition is analyzed using a chromatographic analyzer, and the curve information is recorded based on its position in the formation depth.

The data typically originate from measurements that assess the

components and content of natural gas present in the drilling fluid. During gas testing operations at the drilling site, conventional logging curves are often integrated with gas logging curves. This combination serves to extract the response characteristics and correlational relationships of the layers. It helps evaluating fluid properties and subsequently contributes to the development of a logging data evaluation model.

## 2.2. Characteristics of gas logging curves

The gas content in oil and gas wells is typically categorized into various hydrocarbon components and non-hydrocarbon gas content. Specific hydrocarbon gases include methane, ethane, propane, isobutane, n-butane, isopentane, and n-pentane, among others. Non-hydrocarbon gases primarily consist of nitrogen, carbon dioxide, carbon monoxide, hydrogen, and hydrogen sulfide.

Logging curves differ from conventional discrete sequence curves, time-series curves, and mechanical motion curves in several ways. They exhibit unique characteristics such as concentrated peaks, ambiguous periodicity, low continuity, and abrupt changes. As a result, logging data from different regions require extensive preprocessing. Subsequent research on fluid identification models involves techniques like clustering, image segmentation, sensitive parameter identification, and multi-classification algorithms.

## 2.3. Logging curves

Downhole instruments typically record a data set comprising over 20 sampling curves. These include lithology indicators such as CAL (Caliper), SP (Spontaneous Potential), and GR (Gamma Ray); triple porosity measures like AC (Acoustic), CNL (Compensated Neutron Log), and DEN (Density); resistivity curves represented by RD (Deep Resistivity), RS (Shallow Resistivity), and RMSF (Microspherical Focused Resistivity); as well as gas logging curves, which include total hydrocarbons (THC), C1 (Methane), C2 (Ethylene), C3 (Propane), iC4 (Isobutane), nC4 (n-Butane), iC5 (Isopentane), nC5 (n-Pentane), and so on. The dataset is obtained from 12 offshore natural gas wells in north

China, which is highly unbalanced in fluid types.

## 3. The methodology

In this work, the framework developed for fluid identification comprises three main modules: Module for curve selection based on curve response characteristics and correlational relationships, module for interpretable layer pickup, and module for the feature extraction and classification based on multi-head attention. Fig. 1 illustrates the corresponding flowchart. First, all curves undergo normalization and are input into the 1D-SAX algorithm, resulting in a set of symbols representing the mean and trend changes of the data. Subsequently, the module employs a word frequency algorithm to quantify the differences in each type of symbol across different fluids and compares the degree of difference between the curves. This comparison enables the screening of low-similarity curves, thereby reducing the feature dimensions of the input set of curves.

Following this, the response characteristics and similarity ranking table generated by the previous module are used to screen out a certain number of sensitive curves. These curves are then input into the UMAP algorithm module. This module leverages visualization through dimensionality reduction and clustering to extract and remove high-confidence non-reservoir terms in the feature space, ensuring the balance of data sample categories. Finally, the numerical curves, along with lithological discrete features, are encoded and input into the Transformer module. This module, integrating multiple attention mechanisms, performs fluid classification. The original samples are divided into training, validation, and testing sets, and the optimal model parameter solutions are sought through multiple iterations based on the magnitude of the loss function.

In the following sections, the technical details of each module are described in detail.

## 4. SAX of logging curves

From a feature extraction standpoint, many redundant features exist among all the logging curves mentioned above, and each curve responds

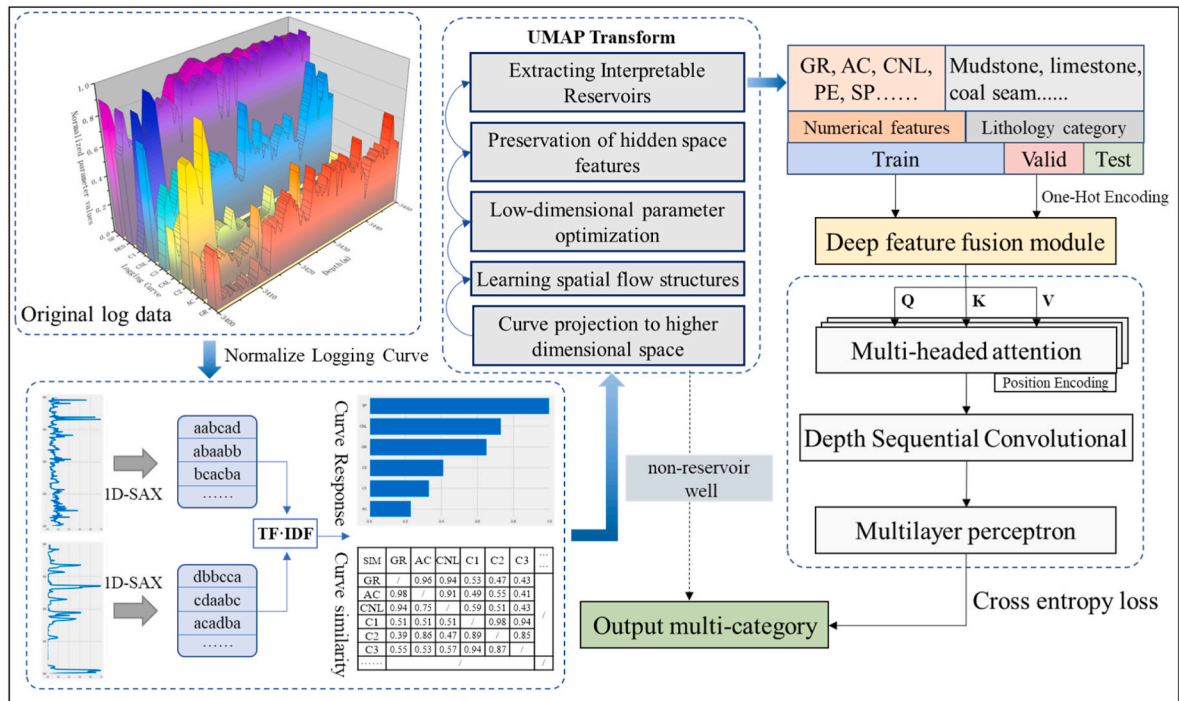


Fig. 1. Flow Chart for fluid identification.

differently to gas and water layers. Conducting sensitivity analysis on these curves before model construction streamlines the subsequent training process. In this section, a sensitivity analysis method combining SAX and term frequency-inverse document frequency (TF-IDF) is presented. SAX is employed to convert curves into symbolic representations, preserving trend changes while compressing logging sequence data. The statistical method based on TF-IDF allows for the evaluation of the importance and relevance of a symbol set in the entire document.

#### 4.1. 1D-SAX representation for logging curves

Among the various methods for extracting features from continuous series through dimensionality reduction, SAX performs symbolic approximation of the data by dividing the time series data into several intervals and averaging each interval (Lin et al., 2007). However, a limitation of this algorithm is its inability to distinguish between upward and downward trends in sampling points when two curves have closely aligned values. To address this issue and capture trend information without increasing computational complexity, a slope parameter is introduced. This addition enhances the accuracy of sensitivity analysis for logging curves.

Each logging curve sequence, characterized by continuous depth attributes, can be represented as  $H = (h_1, h_2, \dots, h_n)$ , where  $n$  is the length of the logging curve to be calculated. The fundamental conversion process of 1D-SAX for each layer value is outlined as follows:

- (1) The sequence is partitioned into sub-segments of length  $l$ . The essence of the SAX algorithm lies in the Piecewise Aggregate Approximation (PAA) algorithm. According to the lower bound measure theorem (Keogh et al., 2001; Lkhagva et al., 2006), data conforming to the confidence interval are designated as symbols in the space and are segmented into multiple truncation regions within that space. The formula for dimensionality reduction is given in the following equation.

$$\bar{h}_i = \frac{w}{n} \sum_{j=\frac{n}{w}(i-1)+1}^{\frac{n}{w}i} h_j \quad (1)$$

where  $n$  is the length of original logging curve, which is segmented into  $w$  subsequences of equal length. Each curve undergoes well logging depth correction, ensuring that the curves have the same length. However, when  $n$  cannot be evenly divided by  $w$ , the length of  $n$  can be fine-tuned through cropping, or a sliding window can be created with a custom subsequence length  $l$  to obtain values sequentially. For instance, if we want to divide a sequence of length 10 into three subsequence segments, we can simply set a sliding window with a width of 4 and a stride of 3 to ensure that each subsequence segment has the same length.

The average sample values of these subsequences are computed, forming the collection  $\bar{H} = \{\bar{h}_1, \bar{h}_2, \dots, \bar{h}_w\}$ . By defining an interval range, the mean set is utilized as a segmented aggregation approximation of the sequence data, while also determining the split point locations based on the interval divisions (Ruan et al., 2021). Table 1 lists the breakpoints with the number of letters from 3 to 6 satisfying the normal distribution, and the breakpoints are used to divide the distribution space into multiple equal probability regions. The series of split points  $\beta$  of the interval

**Table 1**

Lookup table from breakpoints with alphabet sizes from 3 to 6.

| $\beta_i$ | 3     | 4     | 5     | 6     |
|-----------|-------|-------|-------|-------|
| $\beta_1$ | -0.43 | -0.67 | -0.84 | -0.97 |
| $\beta_2$ | 0.43  | 0     | -0.25 | -0.43 |
| $\beta_3$ |       | 0.67  | 0.25  | 0     |
| $\beta_4$ |       |       | 0.84  | 0.43  |
| $\beta_5$ |       |       |       | 0.97  |

are found in the breakpoint lookup table according to the alphabet size, and the mean is expressed as the corresponding letter. Tuning parameters like alphabet size and word length impacts the granularity of representation, influencing the model's ability to capture intricate patterns.

- (2) The depth sequence of each interval is denoted as the average gradient. After the processing of the above PAA algorithm, we can get multiple sub-sequences, and the fluctuation of the trend value of these sub-sequences can obviously reflect the change of stratigraphic properties. Assuming that a certain depth subsequence interval  $[d_1, \dots, d_m]$  is taken, and the mean sign value in this sequence is  $[s_1, s_2, \dots, s_m]$ , the corresponding trend function can be expressed as  $f(x) = \alpha x + b$ :

$$\alpha = \frac{\sum_{i=1}^m d_i s_i - m \bar{D} \bar{S}}{\sum_{i=1}^m d_i^2 - m \bar{D}^2} \quad (2)$$

$$b = \bar{S} - \alpha \bar{D} \quad (3)$$

where  $\bar{S}$  and  $\bar{D}$  are the mean values of  $s$  and  $d$ , respectively, i.e., the curvilinear trend is represented by downscaling to each subsequence according to the least squares method.

- (3) A new sequence is generated based on the symbol table. After each interval undergoes discretization by the PAA value, both the mean and trend indicators are preserved using long strings. Suppose the sequence is divided into four symbol sets, with the breakpoint probability of each symbol following a normal distribution. Fig. 2 illustrates an example of feature extraction from two feature curves utilizing 1D-SAX. Specifically, it demonstrates the implementation results of the AC and C1 curves after Piecewise Aggregate Approximation (PAA) and symbol transformation in the depth interval of 3500 m–3540 m. With a sampling interval of 0.1m, the sequence length in the figure is 400, and the new sequence length after dimensionality reduction is 10, resulting in four mean set regions. At positions where the mean symbols are the same, it becomes challenging to discern curve information. Therefore, the added trend terms are used to reflect the similarity in slope between the two curves.

- (4) The correlation is analyzed based on the generated symbol sequence. A sliding window of width  $L$  is employed to convert the curve into a symbolic representation by depth. The trend value adopts 4 levels as the dimensionality reduction target value, with an interval of  $45^\circ$ . In Fig. 2, the mean word bag of the AC curve from 3500 to 3400 m is represented as "CCDBABAB", and the trend word bag is denoted as "bccdcbbb". Discretization minimizes computational overhead and mitigates the calculation error of sensitivity parameters due to inconsistent peak times when the curve undergoes changes in the formation. The frequency of symbols for each curve is counted, and a word frequency matrix is constructed using the TF-IDF (term frequency-inverse document frequency) algorithm (Alòs et al., 2023). Let  $n_i$  denote the number of the ' $k$ '-th word in the symbolized set, and  $S_{i,k}$  represent the number of the ' $k$ '-th word in the formation label of class  $i$ . For each logging curve, the weight vector  $V$  is calculated using the formula:

$$V = \log\left(\frac{N}{N_i}\right) \times \log(1 + S_{i,k}) \quad (4)$$

where  $N$  is the number of symbol collection categories contained in the entire corpus and  $N_i$  is the number of curves that contain the word. The assumption that exists in this formula is that the words that are most meaningful for distinguishing differences in the curves should be those



that occur more frequently in the symbol collection and less frequently in other documents throughout the large collection of logging curves.

#### 4.2. Correlation and responsiveness analysis

Upon generating the word frequency matrix for each curve, all sets are consolidated as depicted in Table 2. Subsequently, the mean or trend similarity distance between the curves can be computed through summation. Additionally, varying weights can be assigned based on the requirements of different reservoir categories to minimize the redundancy of input features in the ensuing network training. TF-IDF is a numerical statistic used in information retrieval and text mining to evaluate the importance of a term in a document relative to its occurrence in a collection of documents (Zhang et al., 2011). Term Frequency (TF) measures how often a single word vector occurs in a document. It is calculated as the ratio of the number of occurrences of the string in the bag of words to the total number of terms in that bag:

$$TF(t, d) = \frac{\text{Number of times term } t \text{ appears in document}}{\text{Total number of terms in document}} \quad (5)$$

Inverse Document Frequency (IDF) measures the importance of a term in the overall document collection. It is calculated as the logarithm of the ratio of the total number of bag-of-words to the number of bag-of-words containing the term:

$$IDF(t, D) = \log \left( \frac{\text{Total number of documents in the collection}}{\text{Number of documents containing term}} \right) \quad (6)$$

Regarding the response level of an individual curve to various fluid category labels, the relative magnitude of curve morphological alterations across different fluids can be discerned by examining the context of each segment in the string, given the existence of a distance between individual symbols. Sensitivity detection based on multivariate logging parameters can categorize each characteristic curve as a subsystem for anomaly detection. Conducting unit parameter anomaly detection is the basis of multivariate parameter detection, while the key to multivariate parameter anomaly detection is the selection and fusion of feature parameters. The distance measurement method based on TF-IDF integrating trend features effectively improves the efficiency of similarity calculation and meets lower bound requirements. The features of each segment are obtained according to the frequency of symbols, providing a more reliable symbolic expression method for logging sequence data mining.

### 5. The Transformer Classification Method Integrated with UMAP

#### 5.1. The UMAP method

In the interpretation of logging and gas testing data, it is common that the proportion of non-reservoir labels often equals the sum of gas, water layers and all other layers. For fluid identification networks utilizing autoencoder deep learning, an imbalance in data categories can readily lead to a decrease in the recall rate of small sample labels, such as

those representing gas layers. Employing the method of oversampling synthetic minority classes directly for data balance may result in the loss of profound sequence information. Hence, ensuring that the estimator operates within the formation depth range significant for fluid interpretation might hold more importance than directly predicting values. In addition to this, given the complexity of the Transformer architecture, the UMAP algorithm in this section is able to segment the curves, utilizing the non-reservoir layer as a breakpoint, so that the attentional training model in the follow-up only needs to focus on each small segment of the sequence, rather than considering the numerical changes on the entire curve.

The lack of abundant labeled gas layer data and the challenge of model overfitting have persistently been obstacles in applying machine learning to fluid identification (Guo et al., 2018). In this experiment, the UMAP (Uniform Manifold Approximation and Projection) algorithm is used for preliminary statistical analysis of well logging data. Utilizing a semi-supervised machine learning approach, it under-samples certain high-confidence non-reservoir layers. This semi-supervised algorithm is capable of leveraging the available limited labeled sample data and also demonstrates notable generalization capabilities for unknown anomalous input samples.

UMAP, an efficient non-linear dimensionality reduction method, leverages Riemannian geometry and algebraic topology to discover low-dimensional embeddings of structured data (Sainburg et al., 2021). This algorithm is similar to the traditional "graphical segmentation method" in that it produces visualizations of two-dimensional or three-dimensional coordinate graphs. It classifies by partitioning area regions without the need to establish complex assumptions or prerequisites for logging interpretation. The UMAP algorithm consists of two main steps: (1) Approximating the fuzzy structure of high-dimensional datasets using local manifold approximation, and (2) Constructing a loss function and utilizing stochastic gradient descent to ensure high similarity relationships between high-dimensional and low-dimensional samples. The probability relationship between high-dimensional sample points can be articulated as the probability of occurrence of sample point  $i$  given the knowledge of sample point  $j$ . Let  $X = \{x_i, \dots, x_n\}$  be the input data set, which is expressed as:

$$p_{ij} = e^{\frac{d(x_i, x_j) - \rho_i}{\sigma_i}} \quad (7)$$

where  $d$  is the Euclidean distance between the two variables, and  $\rho_i$  representing the minimum distance between the target point  $i$  and the nearest neighbor points,  $\sigma_i$  is the sample constraint factor,  $p_{ij}$  is exponential probability distribution in high dimensions. If only  $k$  nearest neighbor points around high-dimensional points are considered, it can be obtained:

$$\sum_i p_{ij} = \log_2 k \quad (8)$$

In equations (7) and (8),  $\rho_i$  ensures that at least 1 data point with edge weight 1 is selected to be connected to  $x_i$ ;  $\sigma_i$  is the length scale

**Table 2**

Logging curve word frequency matrix. The corresponding mean and trend factors taken in the layer segments for each fluid property and the associated probability of occurrence of these terms are counted.

| Reservoir Category | Mean Factors | Curve I | Curve II | ... | Trend Factors | Curve I | Curve II | ... |
|--------------------|--------------|---------|----------|-----|---------------|---------|----------|-----|
| Gas                | aabbc        | 0.190   | 0.141    | ... | acabb         | 0.081   | 0.087    | ... |
| Gas                | abcde        | 0.168   | 0.000    | ... | bedac         | 0.149   | 0.000    | ... |
| Gas                | ddbcb        | 0.096   | 0.144    | ... | bcabb         | 0.108   | 0.069    | ... |
| Gas                | acabb        | 0.000   | 0.167    | ... | cabaa         | 0.000   | 0.054    | ... |
| ...                | ...          | ...     | ...      | ... | ...           | ...     | ...      | ... |
| Water              | bcaca        | 0.182   | 0.020    | ... | caacb         | 0.062   | 0.054    | ... |
| Water              | bcbaa        | 0.164   | 0.000    | ... | abcac         | 0.139   | 0.000    | ... |
| Water              | cabda        | 0.078   | 0.051    | ... | bcaab         | 0.079   | 0.016    | ... |
| Water              | bcaab        | 0.000   | 0.188    | ... | bbccb         | 0.000   | 0.044    | ... |

parameter. By determining the value of  $\sigma_i$  through the above formula,  $p_{ij}$  is converted into a joint probability matrix with symmetry:

$$p_{ij} = p_{ij} + p_{ji} - p_{ij}p_{ji} \quad (9)$$

For low-dimensional sample points,  $\alpha$  and  $\beta$  parameters are used to regulate the degree of data convergence after mapping from higher dimensions:

$$q_{ij} = \left[ 1 + \alpha (y_i - y_j)^{2\beta} \right]^{-1} \quad (10)$$

The loss between  $p_{ij}$  and  $q_{ij}$  is measured using a binary cross entropy function:

$$CE(X, Y) = \sum_i \sum_j \left[ p_{ij}(X) \log \left( \frac{p_{ij}(X)}{q_{ij}(Y)} \right) + (1 - p_{ij}(X)) \log \left( \frac{1 - p_{ij}(X)}{1 - q_{ij}(Y)} \right) \right] \quad (11)$$

In equation (11),  $X$  and  $Y$  denote the distances between two points in the high-dimensional and low-dimensional sample spaces, respectively. By establishing a soft threshold, the local structure of similar points in the original high-dimensional space that meet the criteria is compressed. This modification regulates the degree of separation between points in the reduced-dimension representation, ensuring that the resultant projection optimally fulfills the segmentation requirements of multiple clusters.

### 5.2. Retaining latent space features

By mapping the original well-logging sequence data to a high-dimensional discrete space, reconstruction can be employed to cluster non-reservoir points that exhibit insensitivity to response changes. By preserving the latent space information of known embeddings and transferring it to new unlabeled data, the UMAP is capable of extracting fluid features that hold significant information. By adjusting the positions of points in the dimensionality-reduced space using scaling, the clustering of data label points is achieved. In equation (11),  $p_{ij}$  applies a local, self-adjustable index kernel to each data point  $x_i$ . Fig. 3 is a schematic representation of high-dimensional point-line attraction clustering, illustrating the construction of a local relationship graph between data sets in a space composed of three features.

Choosing suitable sample distance values and undergoing certain iterations of the gradient descent algorithm enable the control of the topological structure in the reduced latent space. This allows the low-dimensional representation to form multiple cluster partitions when visualized. To precisely capture target values in unknown datasets following data transformation, this study incorporates the K-nearest neighbors algorithm. This algorithm assesses the distance relationships

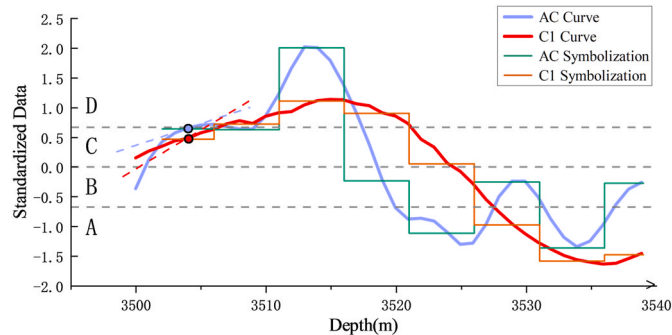


Fig. 2. SAX similarity measure for AC and C1 logging curves. The horizontal coordinate indicates the logging depth in m, and the vertical coordinate is the standardized value of the two curves. At regional points with the same mean, the slope is used to indicate the trend of the curve.

among neighboring points, with the recognition specifically concentrating on non-reservoir sections and establishing a higher aggregation density for clustering non-reservoir labels.

### 5.3. Transformer-based fluid identification framework

Formation depth sequences are subject to various influencing factors, distinguishing them significantly from traditional time series or discrete data classifications and presenting challenges in analysis. The Transformer, originally developed for natural language processing, is a distinctive machine learning network that employs the self-attention mechanism. In tasks related to fluid classification prediction, its performance has the potential to exceed that of traditional deep convolutional or recurrent neural networks in several aspects.

The fluid identification framework in this study, based on the Transformer, adheres to the encoder-decoder concept (Yue et al., 2022). It primarily consists of the feature embedding layer, the transformation layer, and the multi-layer perceptron layer. The transformation layer of the Transformer incorporates deep positional encoding and multi-head attention units. Attention serves as a technique to allocate more weight to specific input data in determining the model's output, concentrating on the most pertinent input data. Feature sequences of interpretative significance from different well-logging curves are segmented based on depth and fed into the embedding layer, with depth information encoding appended to each vector. To bolster prediction accuracy and mitigate overfitting risk, the current input depth layer associates only with a few adjacent layers, eliminating the need to reference the original vector. Unlike existing RNN models for fluid identification (e.g., Vanilla RNN, GRU, LSTM), the Transformer does not process input feature vectors sequentially. It evaluates and discerns the relationships of preceding and succeeding data concurrently, learning through the interconnections between inputs and outputs.

Fig. 4 depicts the workflow of the fluid identification model. The input  $X$  includes continuous numerical features  $x_{num}$  and lithology classification features  $x_{facies}$ . For the former, the value of the characteristic curve is changed with high probability at each depth point. Conversely, the lithology features may remain stable over extended depth intervals. The feature tokenizer converts the input features  $X$  into an embedding vector space  $H \in \mathbb{R}^{k \times d}$  with a dimension of  $d$ ,

$$H = \begin{bmatrix} x_1 * W_1 + b_1 \\ x_2 * W_2 + b_2 \\ \vdots \\ e_{facies} * W_{facies} + b_{facies} \end{bmatrix}_{k \times d} \quad (12)$$

where  $W$  denotes the initialized weights, and  $b$  signifies the feature bias. For lithology features, an inherent distance relationship between the labels is absent. Imposing an ordinal relationship through index encoding and permitting the model to infer an integer arrangement among categories can result in less-than-optimal performance. Consequently, one-hot encoding is executed for each lithology layer  $x_{facies}$ . The ensuing matrix  $e_{facies}$  represents each category in a binary format in every row.

Upon traversing the feature tokenizer, which amalgamates continuous features with character features, it is imperative to preserve the vertical relationship of the embedded vectors and continuously capture the sequence information of the intervals. A 'trainable' CLS (special classification embedding) position classification encoding marker is incorporated (Devlin et al., 2018). In the ensuing training, the embedding vector situated at the CLS position is conveyed to the multi-layer perceptron, namely  $H_0 = [CLS, H]$  (13)

Logging data processed with position coding allows the module to more accurately capture features of localized substructures of the target. As shown in Fig. 5, using layer segments as a sample can address the shortcomings of traditional classification that does not take into account

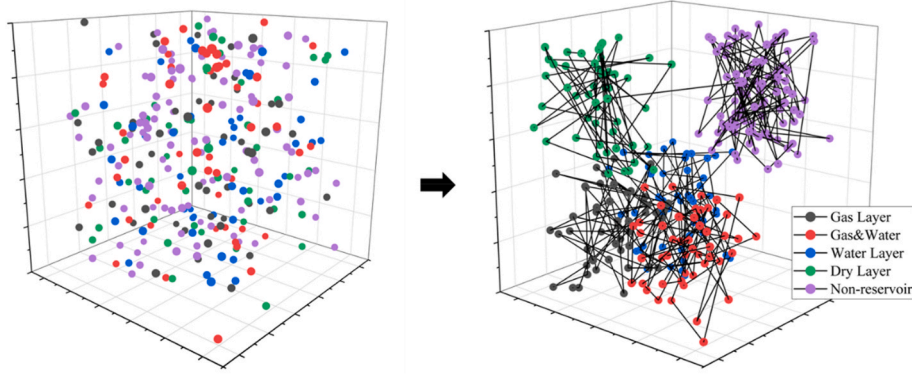


Fig. 3. Adjustment of high-dimensional point-line clustering.

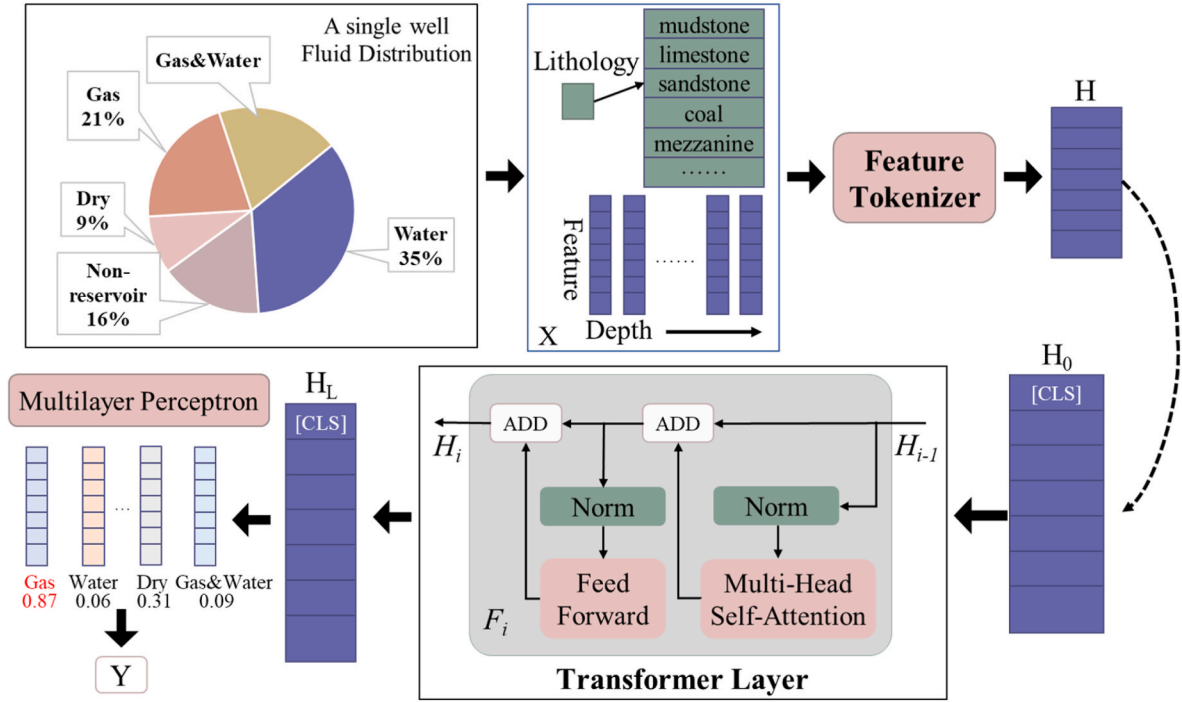


Fig. 4. Fluid Identification framework based on the Transformer.

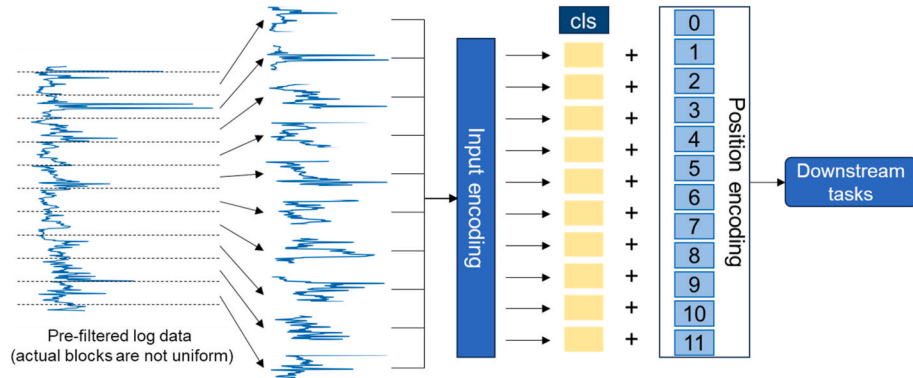


Fig. 5. Relative position coding.

the before-and-after relationship, and pre-screening out the poor reservoirs ensures that the predicted target category has a sufficient share in the training set.

After parameterized encoding, the vector  $H_0$  is batched and fed into multiple Transformer layers. Each layer, represented by the function  $F_i$ , extracts features from the input vector  $[e_{p1}(x_1), e_{p2}(x_2), \dots, e_{pm}(x_m)]$  and

returns a matrix  $H_L$  that has undergone vector cross-embedding. This vector, after undergoing feature attention reconstruction, serves two main purposes. First, it captures the dependency characteristics inherent in the sequence order. Second, it focuses on the cross-dimensional dependency characteristics between different logging curve variables. Finally, the matrix  $H_L$  is input into a multi-layer perceptron (MLP). The number of neurons and layers in this fully connected model are adjusted based on experimental results. Through iterative computation, the cross-entropy loss function is minimized. After obtaining the probability distribution for multi-classification, the *ReLU* activation function is used to output the predicted fluid results.

#### 5.4. Multi-head attention mechanism unit

The core of the Transformer layer is formed by stacking multi-head attention units in multiple layers. These units enable the model to transmit attention parameters across different depth positions, extracting relationships between vectors originating from disparate layers. Subsequently, this information is conveyed to the succeeding layer during feed-forward propagation, allowing the capture of more enriched stratigraphic information. As shown in the Transformer layer in Fig. 4, the data is normalized before the multi-head attention module and the fully connected network to prevent gradient explosion or gradient vanishing in the model. A self-attention layer fundamentally comprises three distinct parameter matrices:  $Q$  (Query),  $K$  (Key), and  $V$  (Value). When these matrices are multiplied with the input vector  $x_i$ , the corresponding weights are derived. Firstly,  $x_i$  is simply mapped into three linear mappings,  $Q = x_i W_q$ ,  $K = x_i W_k$ , and  $V = x_i W_v$ , to obtain the  $d_Q$ -dimensional query matrix, the  $d_K$ -dimensional key matrix, and the  $d_V$ -dimensional value matrix, and the attentional weights are obtained by calculating the product of the query and key matrices and dividing it by the scaling factor, and then applying the activation function to obtain the attentional weights. The self-attention output is obtained by weighting the value matrix  $V$  according to the values of the attention weights. The method of computation is outlined below:

$$\text{Attention}(Q, K, V) = A \times V = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (14)$$

where  $A$  is the similarity matrix obtained by multiplying the  $QK^T$  matrices. And  $d_k$  represents the dimension of  $k$ , divided by the square root of the dimension of  $k$  to keep the gradient stable during training.

The degree of similarity between two vectors is determined through the dot product. For each embedded vector, the attention matrix  $A$  computes the level of attention given to other embedded vectors, thereby generating attention weight scores that reflect the contextual relationships among the vectors. Considering the extensive length of the well logging curve, inputting the entire data with excessively high dimensions can substantially escalate the sequence complexity of the Transformer. Attention scores typically follow a long-tail distribution, with the majority of the scores being relatively minor. Hence, it is crucial to model only those relationships that are significant when calculating correlations. This optimization can be realized by allowing the key values in the index to form a sparse attention matrix with the query key values, significantly reducing computational burden (Zhou et al., 2021). Multi-head attention, which learns different features of the vector by stacking multiple attention layers, can be represented as follows:

$$\text{MultiHead}(\bar{Q}, \bar{K}, \bar{V}) = \text{Conact}(\text{head}_1, \dots, \text{head}_n)W^O \quad (15)$$

$$\text{head}_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V) \quad (16)$$

where Multi-Head Attention is to do the Scaled Dot-Product Attention (Equ.12) process multiple times and then combine the outputs.

Upon concatenating the attention matrices of  $n$  attention heads, which incorporate the [CLS] encoding classification token, these

matrices are subsequently multiplied by the weight matrix  $W^O$ . This operation yields the feature extraction matrix corresponding to the well logging curve at that specific depth position. The ultimate probability outcomes for fluid multi-classification are attained through the utilization of a Multi-Layer Perceptron (MLP) network.

$$y_{out} = \text{MLP}\left[\text{Linear}\left(\text{ReLU}\left(\text{LayerNorm}\left(H_L^{(CLS)}\right)\right)\right)\right] \quad (17)$$

where a batch normalization is done on the  $H_L$  matrix embedded with the CLS vector header, and an activation function is used to map the inputs of the neurons to the outputs. This matrix is then pooled and passed through the MLP network to output the specific prediction probabilities for each label.

When each stratum is sequentially input into the model, a "jump phenomenon" may manifest in the predicted layers. This implies that, despite the overall high accuracy in fluid identification, there might be sporadic instances where labels undergo abrupt changes within a confined range in specific strata. For example, a water layer might momentarily emerge within a gas layer, only to swiftly revert back to being a gas layer, a scenario that contradicts actual geological conditions.

Fig. 6 clearly shows that the process of achieving fluid identification results involves operations such as segmentation, concatenation, fusion, convolution, and iterative optimization on the original dataset. These complex transformations may result in differences between the tail-end data and the original head data, potentially introducing unknown perturbations in the predictions for certain sensitive positions, leading to abrupt changes in the predicted geological formations that do not align with the target outcomes.

Such transient alterations are attributable to the presence of sample noise or a disruption in regularity. To address this, change positions are marked with a binary label (0–1). If changes are observed more than twice within a predetermined depth threshold, these anomalies are rectified using a neighboring point interpolation method. This approach not only mitigates sudden fluctuations and augments data smoothness but also yields a slight enhancement in recognition outcomes.

## 6. Experiments and results analysis

### 6.1. Data overview and pre-processing

The logging data for this research comes from the interpretation results of a gas reservoir production area in offshore China. It integrates logging curves for 27 technical parameters, including conventional logging data (such as Gamma Ray, triple porosity, resistivity), lithology, and gas logging Curves (including light hydrocarbons and non-hydrocarbons). Twelve wells with relatively complete data samples from this area were selected. Based on the logging interpretation and gas testing results provided by the production site, the fluids were categorized into dry layers, gas layers, gas-water co-layers, water layers, and non-reservoir layers with no interpretative significance. The sampling interval for the training data is uniformly set at 0.1m, starting from 3000m and ending between 3600m and 3800m. Since the changes in reservoir properties have a relatively stable upward and downward trend, logging parameters usually do not undergo multiple drastic changes within a narrow depth range. Therefore, a few missing values can be filled using the nearest neighbor interpolation method. Finally, the data is standardized and feature-scaled to distribute within a space with a mean of 0 and a standard deviation of 1.

### 6.2. Sensitive curve selection and processing

Selecting curves that significantly impact the results can enhance the model's recognition capability. In this study, we employed the symbolic trend approximation aggregation method to analyze the curves. Initially, we computed and stored the symbolic vector representation of



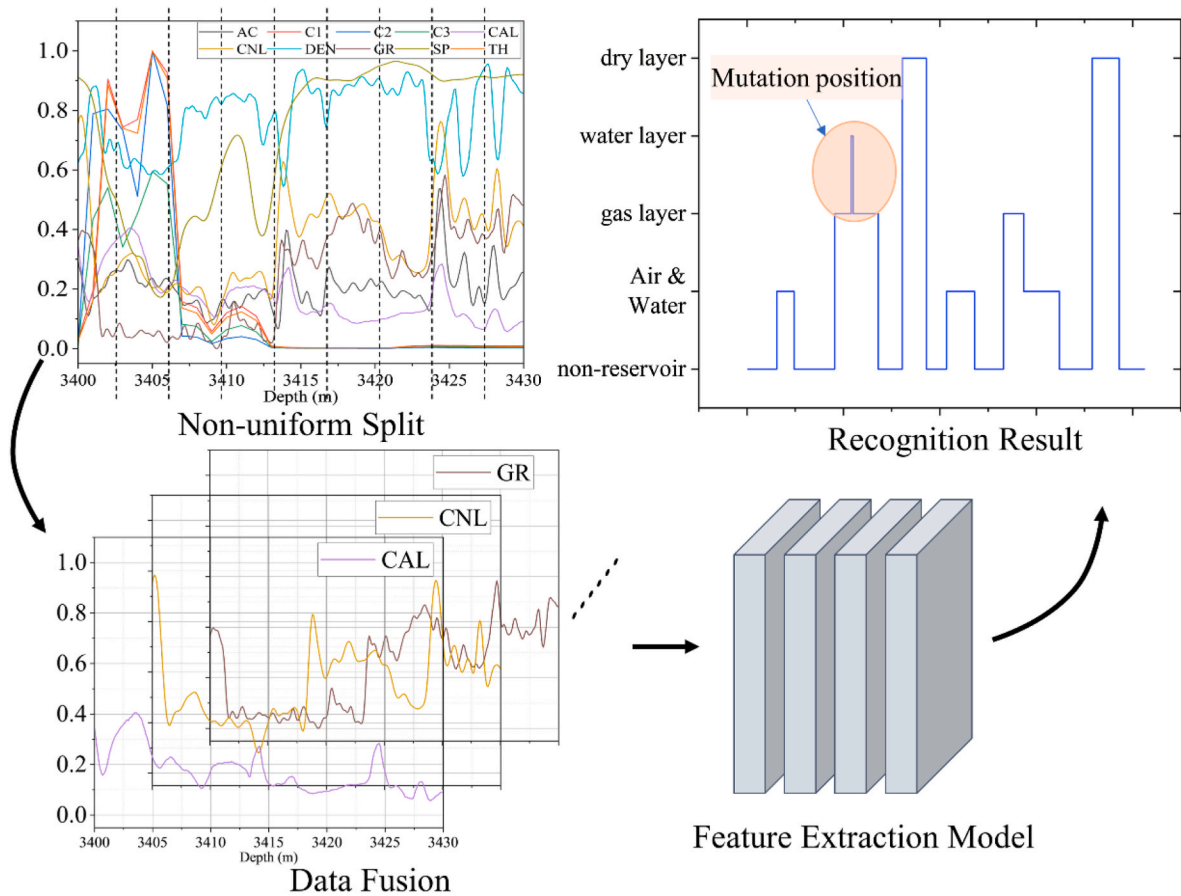


Fig. 6. Basic Procedure to obtain fluid identification results.

the "mean term" and "trend term" of the curve values. A sliding window was used to segment and extract these terms, forming a vector word bag.

For the sensitivity ranking calculation of the curves, only the changes within each curve were considered. The objective was to identify feature curves that either exhibit pronounced response characteristics in different reservoirs or maintain high values. When there's a change in the reservoir fluid within a certain range of a layer segment, a secondary sampling is performed. These samples are then added to the word vector collection, summed up, and the final value was output.

Fig. 7 depicts line charts of well logging curves and their corresponding derived response features for four different fluid types. The computed correlations between fluid types and curve characteristics are

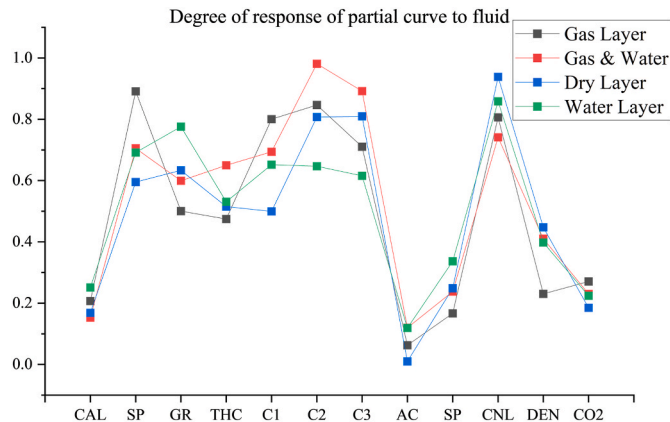


Fig. 7. Response curves to four types of fluid contents. The horizontal axis shows different logging curves, while the vertical axis shows sensitivity.

represented on a scale from 0 to 1. It can be observed from the figure that there is not a significant difference in response characteristics for fluids with different properties. Hydrocarbon content is an important parameter for determining fluid properties, while the resistivity curve, sensitive to fluids, may not be as effective. In the study area, there is a phenomenon where the resistivity of water layers is close to or even higher than the resistivity of gas layers.

For modeling the correlation of curves, the discrete word symbol bags are weighted using the TF-IDF algorithm and then transformed into feature weight vectors. The cosine similarity of the feature vectors is calculated pairwise for all curves. After summing, they are normalized to uniformly determine their similarity. Fig. 8 presents the correlation matrix of the curves. By comparing these with the response values of different reservoir curves, a few optimal feature curves are selected between the two to serve as training samples for the input layer.

### 6.3. Training process and sample results

In this experiment, we used a single well as a case study. The optimal curve combinations GR, SP, C1, AC, CNL, RD, and lithology classification for the interval were selected as the training features. During fluid prediction, the spatiotemporal dimensional features between the curves were calculated. This means that not only the internal sequential relationship of the curve itself was considered, but the interrelationships of multiple curves at each layer were also emphasized for prediction.

Initially, using the UMAP method, some non-reservoir structures with prominent features were excluded. This was done to avoid using data balancing algorithms that overly preprocess the original data, thereby reducing the interference of reservoirs without interpretative significance on the classifier. By adjusting the dimensionality size after dimensionality reduction, the optimal reservoir partitioning space

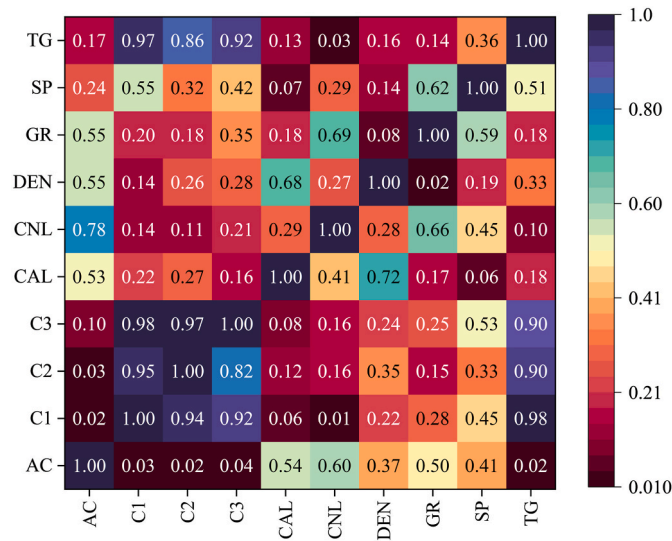


Fig. 8. Correlation matrix of partial curves. The color scale represents correlation.

was sought. The main parameters of UMAP include the number of neighboring points calculated and the minimum low-dimensional distance between two points. The former represents the algorithm's focus on the local or overall structure of the data, while the latter controls the packing density of the points after dimensionality reduction.

The data can be dimensionally reduced by calling the input dataset and setting the hyperparameters (McInnes et al., 2020). The parameter values are determined by a grid search.  $n\_neighbors$  is the most important hyperparameter, which is used to specify the number of approximate nearest neighbors. This parameter controls the UMAP algorithm to balance the local and global structures, with larger values focusing on the global structure and smaller values focusing on the local structure, and is recommended to be set to 5–8.  $min\_dist$  represents the minimum distance between the samples in the low-dimensional space. This parameter controls how closely the UMAP algorithm clusters the points together; larger values result in dispersed samples; set the parameter to 0.3. The metric parameter defines the Euclidean distance used by the UMAP algorithm to measure the distance along the flow pattern.

Principal Component Analysis (PCA) is a traditional linear method for dimensionality reduction (Maćkiewicz and Ratajczak, 1993), which aims to find a low-dimensional space that preserves the important structure of the data. In models involving multiple data types, the PCA algorithm is often used to capture features, such as in anomaly detection of drilling data time series (Altindal et al., 2024). However, linear algorithms such as PCA cannot deal with the nonlinear mapping from high-dimensional data to low-dimensional data, and are not suitable for extracting features with highly correlated parameters, and the fluid types and gas content of formation rocks often have complex nonlinear relationships. Compared with the UMAP algorithm, PCA algorithm's accuracy for the identification of non-reservoir structures on this dataset is 8.6% lower.

Fig. 9 shows the scatter plot visualizing the data placed in a three-dimensional space. It can be observed that the classification difficulty for the reservoir is relatively high, making it challenging to form clusters through clustering. However, due to the nature of the non-reservoir, it can be clearly grouped after dimensionality reduction. Prioritizing the accuracy of non-reservoir label recognition, the KNN (k-Nearest Neighbor) algorithm was used with a lower number of neighboring points. The points of the non-reservoir that are close in spatial distance were removed and input into the next layer of the trainer.

After processing with the UMAP algorithm, the raw logging data are divided into multiple layer fragments focusing on the reservoir, which

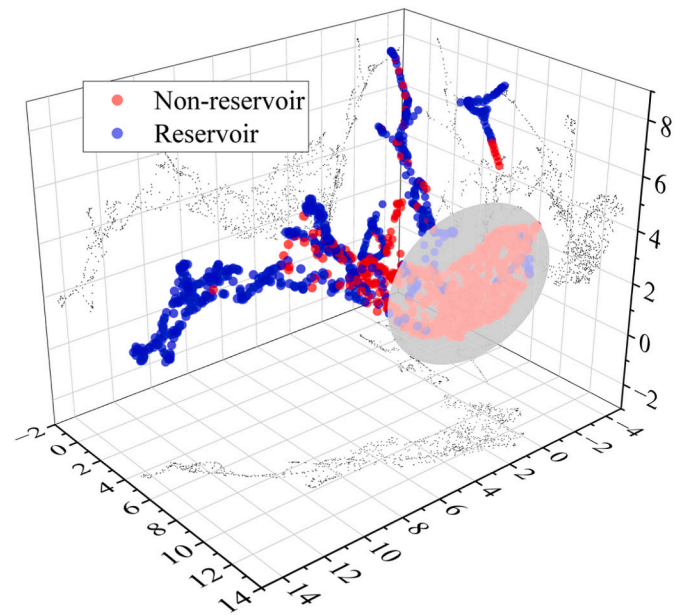


Fig. 9. Reservoir division based on UMAP.

greatly reduces the length of the sequence and allows the Transformer to perform before-and-after causal inference with less memory consumption. The gas logging data used in this paper were sampled at a depth interval of 1 m, with a total of 201,000 sample points from 335 data sets. The remaining 10,793 sample points entered the attention network after pre-screening by the algorithm described above. Because continuity of the data is essential, a set of depth layer segment bindings is used as input, with training, test and validation data ratio of 6:2:2 and repeated random sampling for model validation. To mitigate the impact of excessively long input vectors on the results during training and prediction, a sliding window is set up. Using non-reservoirs and a fixed maximum value as breakpoints, unevenly spaced different depth stratigraphic intervals are segmented. After position vector encoding, these segments are batch-input into the Transformer-based classifier to capture and analyze global features. To mitigate the impact of excessively long input vectors on the results during training and prediction, a sliding window is set up. Using non-reservoirs and a fixed maximum value as breakpoints, unevenly spaced different depth stratigraphic intervals are segmented. After position vector encoding, these segments are batch-input into the Transformer-based classifier to capture and analyze global features.

Within the Transformer, the hyperparameters are initially set based on a priori knowledge, then gradually adjusted for multiple experiments to assess the stability and consistency of the hyperparameters in different runs. The multi-head attention mechanism is configured with 5 heads. The learning rate is set at 0.01, with an iteration count of 500. Three fully connected layers and a tanh activation function are used for final identification. If the loss function on the validation set does not improve (i.e., decrease) over 10 consecutive iterations, the training is halted prematurely, and the model is output. Regarding the model's recognition bias, Table 3 demonstrates the numerical relationship between the true and predicted values. The classification algorithm using multiple attention heads does not have exhibit a clear preference for a any specific category.

Validating models and assessing their generalization capabilities are common practices in machine learning. One technique for this purpose is K-fold cross-validation, which involves partitioning the dataset into K equal-sized folds or subsets. During each iteration, the model is trained on K-1 folds and validated on the remaining fold. This process is repeated K times, with each fold serving as the validation set once. The performance metrics are then averaged over the K iterations to provide a

**Table 3**

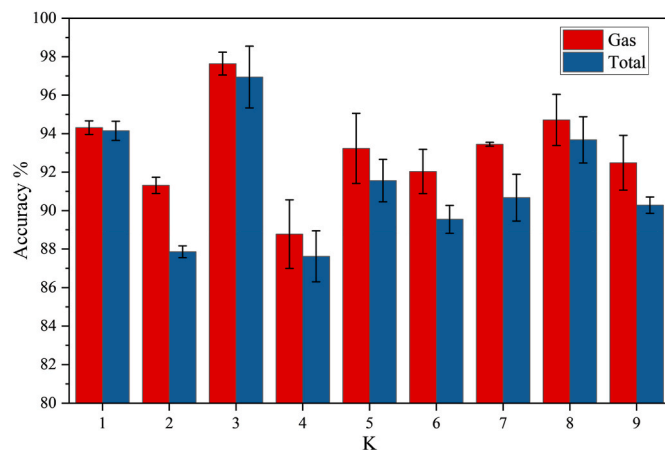
Comparison of different networks. A variety of algorithms are used to predict the gas layer, gas-water co-layer, water layer, dry layer and poor gas layer of the overall data.

| Methods            | Number of Validation Set/Correctly predicted Value |        |         |       |          | Accuracy % |
|--------------------|----------------------------------------------------|--------|---------|-------|----------|------------|
|                    | Gas                                                | G&W    | Water   | Dry   | Poor Gas |            |
| KNN                | 91/81                                              | 111/91 | 355/294 | 52/41 | 93/60    | 80.77      |
| Decision Tree      | 91/82                                              | 111/83 | 355/284 | 52/45 | 93/72    | 80.63      |
| SVM                | 91/74                                              | 111/89 | 355/295 | 52/47 | 93/84    | 83.90      |
| DNN                | 91/79                                              | 111/84 | 355/298 | 52/49 | 93/62    | 81.48      |
| XGBoost            | 91/85                                              | 111/91 | 355/304 | 52/49 | 93/87    | 87.75      |
| UMAP + XGBoost     | 91/86                                              | 111/91 | 355/302 | 52/49 | 93/86    | 87.46      |
| UMAP + Transformer | 91/89                                              | 111/98 | 355/312 | 52/49 | 93/86    | 94.02      |

comprehensive evaluation of the model's predictive power. In the work, the dataset is divided into 10 equal-sized folds, with the model trained on 9 folds and validated on the remaining fold. The accuracy metrics are depicted in Fig. 10, and the standard deviation is utilized for the error bars.

To further validate the fluid identification performance of the proposed model, this study employed the Scikit-Learn and PyTorch machine learning libraries to construct five different algorithmic models: K-Nearest Neighbors (KNN), Decision Tree, Support Vector Machine (SVM), XGBoost, and Deep Neural Networks (DNN) for a specific well in the region. Fig. 11 illustrates the complete training process of the model described in this paper, showing the prediction results in terms of accuracy. The method based on the Transformer network, as described in this paper, demonstrates some improvement compared to traditional machine learning models, fitting the actual gas test results more accurately. Additionally, the method presented in this paper exhibits a high recall rate for gas layers and, when excluding non-reservoir labels, does not overlook critical geological information.

From the prediction results shown in Table 3 and it can be observed that the Transformer classification method integrated with UMAP demonstrates a notably high recognition accuracy rate, outperforming the other models. The KNN and SVM algorithms, which rely on distance computation to categorize data points in space, may have low processing accuracy when dealing with high dimensional data. Decision tree and



**Fig. 10.** K-fold cross validation error bar. It shows identification accuracy for gas layers and all strata classes.

XGBoost integration algorithms may not be flexible enough for continuous-type features, and the internal use of finding the optimal splitting point may lead to the loss of information on continuous values. The Transformer model has a unique ability to express model features due to its ability to focus on the global structural information of the reservoir while focusing on the local structural information of the target layer segments. This shows that the dimensionality reduction capability of UMAP combined with the attention mechanism of the Transformer, can provide a robust solution for fluid identification in the context of logging data.

## 7. Conclusion

This study, focusing on well logging data primarily derived from gas measurement, employs a machine learning approach centered on dimensionality reduction and curve reconstruction. From curve sensitivity analysis to reservoir importance selection, and from feature extraction based on multi-head attention mechanisms to model optimization, a comprehensive fluid identification method has been established. Using this method, a high accuracy rate was achieved in a fluid identification experiment in a certain oil and gas field in offshore China, demonstrating certain advantages over other machine learning methods.

- (1) When faced with the challenge of selecting from a vast array of well logging curves with diverse properties, this study proposes a symbolic bag-of-words approach to indicate curve mean variations and trends. By integrating with text processing algorithms, it evaluates the sensitivity to target fluids and the correlation between curves, providing reliable pre-processed input data for subsequent models.
- (2) The UMAP algorithm is employed for dimensionality reduction and visualization analysis of reservoirs. It allows for a clear visualization of non-reservoirs exhibiting distinct cluster attributes in vector space. By masking the densely aggregated non-reservoir classes in advance, the classifier's recognition performance is significantly optimized.
- (3) The multi-head attention encoder, augmented with positional encoding, ensures that the classification model takes into account two key factors during training. First, it considers the relationship of curve variations near areas where reservoir properties change. Second, it focuses on the numerical relationship changes between feature curves. The recognizer is versatile, capable of processing both continuous numerical curve inputs and discrete lithology labels, thereby achieving a robust multi-scale fluid identification objective.

However, there are also some potential limitations. Firstly, due to limited gas testing operations, the availability of the training dataset is limited, which could affect the accuracy of the prediction model. Secondly, similar to other deep learning, Transformer requires a large amount of memory to store the training iterations, and has poor interpretability, especially for the lack of intuitive representation of attention weights. Finally, in the absence of pre-training, the algorithm may lead to a decline in model performance due to too unbalanced data.

In future studies, the parameterization of the multiple attention will be further investigated to speed up and simplify the parameterization, and further optimization of the curve sensitivity detection is needed. In addition, other Transformer-based tasks will be explored, such as the use of pre-training that is more closely aligned with downstream tasks, and the fusion of continuous and discrete data. In summary, this research offers a holistic approach to fluid identification, integrating advanced machine learning techniques with domain-specific knowledge in petroleum geology, providing a robust solution for the challenges faced in well logging analysis.



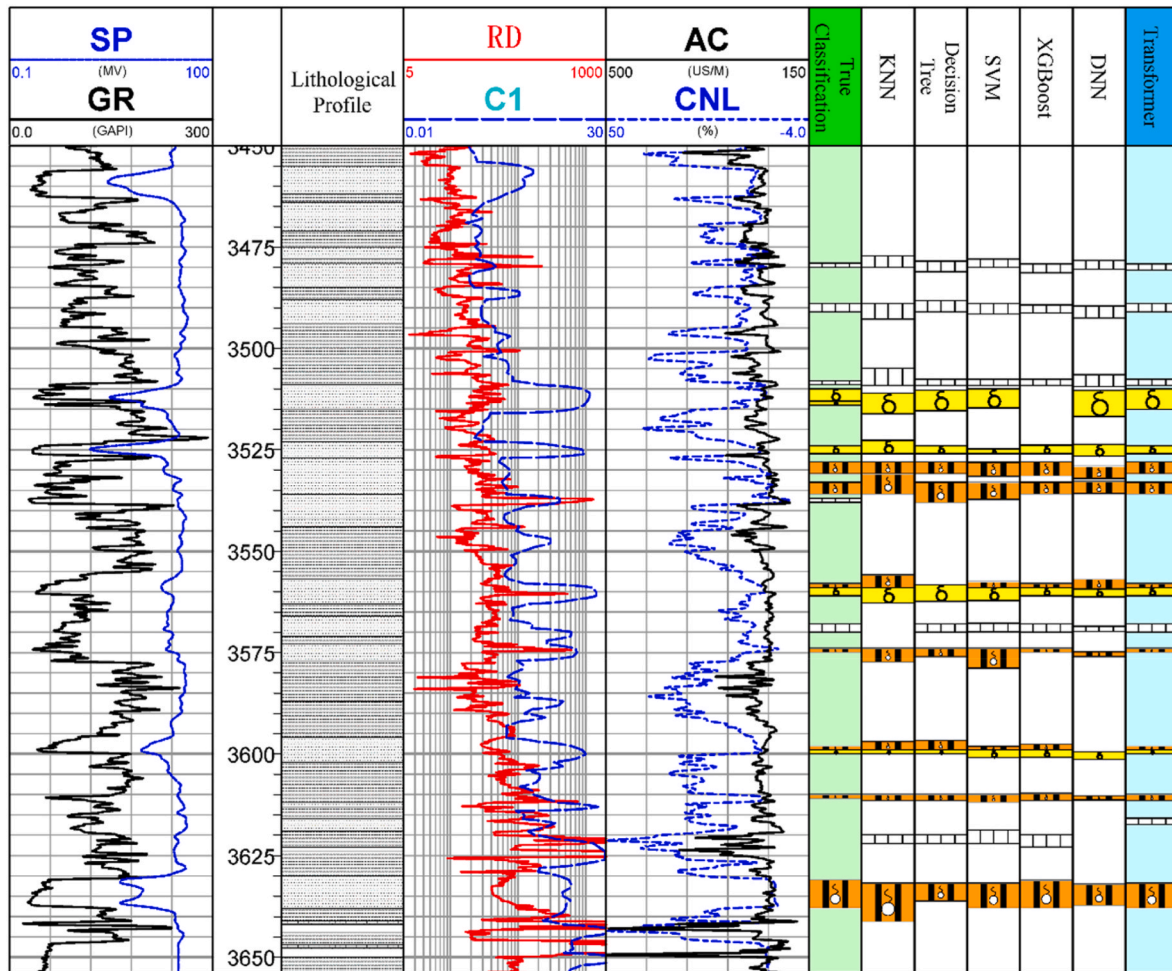


Fig. 11. Comparison of prediction results of different machine learning algorithms for a specific well in the region. Transformer represents the algorithm developed in this paper, five other algorithms include KNN, Decision Tree, SVM, XGBoost and DNN.

#### CRedit authorship contribution statement

**Yuanpeng Hua:** Writing - original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Guozhong Gao:** Writing - review & editing, Validation, Supervision, Software, Resources, Project administration, Methodology, Funding acquisition, Conceptualization. **Daxiang He:** Writing - review & editing, Software, Resources, Methodology, Funding acquisition. **Gang Wang:** Writing - review & editing, Supervision, Resources, Project administration, Methodology. **Wenjun Liu:** Writing - review & editing, Software, Resources, Methodology.

#### Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

#### Data availability

The data that has been used is confidential.

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